Lecture 09 - Correlation and Regression

This lecture moves from studying one variable at a time to studying relationships between variables.

In earlier lectures, the main object was usually a single quantity: an average return, a standard deviation, a distribution, a confidence interval, or a hypothesis about one population mean. Regression begins when the question changes. We no longer ask only what one variable looks like. We ask how two variables move together, and whether one variable helps us understand, summarize, or predict another.

That shift is central in finance and business. A firm may ask whether investment is associated with revenue. A marketing team may ask whether advertising spending helps explain sales. A credit analyst may ask whether leverage helps predict default risk. In each case, the data come in pairs or groups of variables, and the task is to describe the relationship without overinterpreting it.

The lecture has two connected parts. The first part introduces regression as a statistical model and explains fitted values, residuals, least squares, correlation, the coefficient of determination R^2, and Python output. The second part asks when a fitted model is useful for prediction, with special attention to interpolation, extrapolation, and model complexity.

A non-graded optional appendix at the end applies the same regression logic to financial beta.

1. Models as Simplified Descriptions

A model is a simplified mathematical description of reality. It links one or more inputs to an output we want to understand, measure, or predict.

In a teaching example, we may know the rule that generated the data. For instance, we might generate points from a straight line plus noise, then ask whether a regression can recover the pattern. In real applications, the direction is reversed. We observe the data first. Then we look for a model that captures the useful part of the relationship.

The word "useful" is important. A model does not need to reproduce every observation exactly. It needs to summarize the structure that matters for the question. A model used for forecasting may be judged by predictive accuracy. A model used for explanation may be judged by whether the estimated relationship is stable and interpretable. A model used for causal analysis requires a much stronger design.

2. Deterministic and Statistical Models

A deterministic model gives an exact output once the inputs are known. If the model is

```
y=2x,
```

then x=10 gives y=20. There is no unexplained variation left.

Most business and finance problems are not like this. Two customers with similar profiles can spend different amounts. Two firms with similar advertising budgets can generate different sales. Two borrowers with similar income and leverage can still have different credit outcomes.

A statistical model allows for that variation. A simple way to write the idea is

```
y=f(x)+epsilon.
```

The function f(x) is the systematic part of the relationship. The term epsilon captures the part not explained by the variables in the model. This unexplained part may come from omitted variables, measurement error, timing, shocks, or plain randomness.

The goal is not to eliminate noise by pretending it does not exist. The goal is to estimate the average pattern while keeping track of what the model leaves unexplained.

3. Linear Relationships

The simplest regression model is a straight line:

$$\hat{y}=b_0+b_1x.$$

The fitted value $\hat{y}$ is the value predicted by the line. The intercept b_0 is the predicted value when $x=0$. The slope b_1 is the predicted change in y when x increases by one unit.

The slope is usually the main object of interpretation. If revenue is regressed on investment and the slope is 2, then one additional unit of investment is associated with two additional units of predicted revenue, on average in the fitted model.

A straight line is not always appropriate. Some relationships curve, flatten, accelerate, or change direction. This is why the scatterplot comes before the regression output. If the shape of the data is clearly nonlinear, a linear model may be a poor summary even if it is easy to estimate.

4. Inputs, Outputs, and the Question

In regression, the dependent variable is the output we want to explain or predict. The explanatory variable is the input used to do that.

These roles are not fixed by the dataset. They are fixed by the question. Suppose a dataset contains amount spent, age, number of items bought, payment method, and satisfaction. If the question is whether spending predicts satisfaction, satisfaction is the dependent variable. If the question is whether satisfaction predicts future spending, spending becomes the dependent variable.

Simple regression uses one explanatory variable. Multiple regression uses several:

$$\hat{y}_i=b_0+b_1x_{1,i}+b_2x_{2,i}+...+b_kx_{k,i}.$$

The interpretation changes slightly in multiple regression. The coefficient b_j is read as the effect of x_j holding the other variables fixed. That phrase matters because each coefficient is estimated in the presence of the other inputs.

5. Fitted Values, Residuals, and Error

Once a regression line has been estimated, each observation has a fitted value:

$$\hat{y}_i=\hat{b}_0+\hat{b}_1x_i.$$

This fitted value is the model's prediction for observation i. It is usually not equal to the observed value y_i. The difference is the residual:

$$e_i=y_i-\hat{y}_i.$$

A positive residual means the observation is above the fitted line. A negative residual means it is below the fitted line. Residuals are the measured errors after fitting the model.

It is useful to distinguish residuals from the theoretical noise term. The model contains an unobserved error term ϵ_i. After estimating the line, we observe residuals e_i. The residuals are not the true errors, but they are the sample evidence about what the line failed to explain.

Good residuals do not need to be small for every observation. Real data are noisy. What matters is whether the residuals look like unstructured noise or whether they show a remaining pattern. A curve in the residuals suggests that the model may be missing a nonlinear relationship. A widening spread suggests that the variability of the errors may change across the range of fitted values.

6. Least Squares

There are many possible lines through a cloud of points. We need a rule for choosing one.

Ordinary least squares chooses the line that minimizes the sum of squared residuals:

$$\sum_{i=1}^n e_i^2.$$

This means we look at the vertical distance between each observation and the fitted line, square those distances, and choose the line that makes the total as small as possible.

Squaring has two effects. Positive and negative residuals do not cancel out. Large errors also receive more weight than small errors. A residual of 10 contributes much more to the criterion than a residual of 2.

Least squares is not the only possible definition of "best fit." It is the standard starting point because it is simple, widely used, and directly connected to the regression output produced by software.

7. Correlation, r^2, and R^2

Correlation measures the direction and strength of a linear relationship.

A positive correlation means the variables tend to move together. A negative correlation means one tends to be high when the other is low. A correlation near zero means little linear association, although there may still be a nonlinear relationship.

Correlation is standardized. It always lies between -1 and 1. This makes it useful for comparing relationships across pairs of variables measured in different units.

In simple linear regression with an intercept, the coefficient of determination R^2 is connected directly to the squared sample correlation r^2:

$$R^2=r^2.$$

The correlation r keeps both direction and strength. The squared correlation r^2 removes the sign. The coefficient of determination R^2 reports strength of fit only. A strong negative linear relationship can therefore have a high R^2. The sign is read from the correlation or from the slope.

R^2 is a sample fit measure. It tells us how much of the variation in the dependent variable is explained by the fitted linear relationship. It does not prove causality, and it does not guarantee that the model will predict well outside the data used to estimate it.

8. Association and Causation

Correlation and regression describe association. Causation is a stronger claim.

Suppose sales and temperature move together. The scatterplot may show that warmer days are associated with higher camping sales. That association may be useful for planning inventory. It may also fit a plausible story: warmer weather encourages camping.

Still, the graph alone does not prove the mechanism. Maybe warm days coincide with holidays. Maybe advertising increases during summer. Maybe another factor affects both temperature-related behavior and sales.

To make a causal claim, we need more than a fitted line. We need context, timing, controls, institutional knowledge, or an experimental or quasi-experimental design. Regression can be part of causal analysis, but it does not become causal automatically.

9. Reading Regression Output in Python

In Python, ordinary least squares regression can be estimated with `statsmodels`.

The workflow is conceptually simple. Load the data. Choose the explanatory variable or variables. Add a constant so the model estimates an intercept. Choose the dependent variable. Fit the model.

The command `sm.add_constant(X)` adds a column of ones to the explanatory data. This is how the model estimates the intercept. Without that column, the fitted equation is forced through zero:

```
y_hat_i=b_1x_i.
```

With the constant, the model estimates

```
y_hat_i=b_0+b_1x_i.
```

Regression output should be read in pieces. First, identify the dependent variable and the sample size. Then read the coefficients. The constant is the intercept. The coefficient on the explanatory variable is the slope. The output also gives standard errors, t-statistics, p-values, and confidence intervals. These quantities describe uncertainty around the estimated coefficients.

Finally, read R^2 as a goodness-of-fit measure. It tells us how much of the dependent variable's variation is explained by the fitted regression in the sample.

For the official Python exercises, use `data_lecture9_exercises.xlsx`: sheet `data1` for the simple regression example, and sheets `data2`, `data3`, and `data4` for comparing linear fits across different shapes. The separate lecture demonstration of multiple regression uses `data_lecture9.xlsx`, sheet `multivariate`.

10. Covariance, Correlation, and the Slope

It is useful to connect three related quantities: covariance, correlation, and the regression slope.

Covariance measures direction of co-movement in the original units. Positive covariance means two variables tend to be above their averages together and below their averages together. Negative covariance means one tends to be above its average when the other is below.

The limitation is scale. Covariance depends on the units of the variables, so its size is hard to compare across problems.

Correlation standardizes covariance:

```
r=fracCov(X,Y)s_Xs_Y.
```

Dividing by the standard deviations removes the units. The result lies between -1 and 1.

The regression slope uses the same co-movement idea but keeps the units needed for prediction:

```
b_1=fracCov(X,Y)Var(X).
```

Equivalently,

```
b_1=r(s_Y / s_X).
```

This formula is useful because it separates the pieces. Correlation gives standardized direction and strength. The ratio of standard deviations converts that standardized association into units of Y per unit of X. That is why correlation and slope are related, but not the same object.

11. Prediction: Interpolation and Extrapolation

After estimating a regression, it is tempting to use it for prediction. The first question is where the prediction is being made.

Interpolation means predicting inside the range of observed data. If investment values in the sample run from 10 to 25, predicting revenue at investment 18 is interpolation. Nearby observations support the prediction.

Extrapolation means predicting outside the observed range. Predicting revenue at investment 80 uses the fitted relationship in a region where the model has not seen data. This is usually much riskier.

The issue is not that extrapolation is mathematically difficult. The equation can always produce a number. The issue is whether the relationship still holds outside the observed range.

Growth examples make the danger clear. A company with one million followers growing by 20 percent per month can produce an enormous four-year forecast if the growth rate is extended mechanically. The arithmetic may be correct. The model is not realistic if it ignores market size, saturation, competition, and changing behavior.

With many explanatory variables, extrapolation can be hidden. A new observation may look normal one variable at a time, while the full combination of variables is outside the region represented in the sample.

12. Model Complexity and Parsimony

The same data can often be described by many different models.

A simple model is stable. It does not change much when the sample changes. But it may miss important structure. This is underfitting. The model has high bias because its predictions are systematically too crude for the real relationship.

A very flexible model can follow the training data closely. It may even follow noise. This is overfitting. The model has high variance because small changes in the data can lead to a very different fitted model.

The bias-variance tradeoff summarizes this tension. More complexity can reduce bias because the model can represent richer patterns. Too much complexity raises variance because the fitted model becomes sensitive to the particular sample.

Good prediction usually requires a balance. The model should capture the main structure while still performing well on new data. This is why cross-validation, model selection, and regularization matter in applied predictive work.

Parsimony means choosing a model that is simple enough to be stable and interpretable, while still rich enough for the question.

The supplementary Python practice includes a light train/test split using the same simple linear regression model as before. The training sample is the part of the data used to estimate the line. The test sample is held back and used only to check predictions on observations that were not used for fitting. This is an optional programming extension, not part of the examinable slide material.

Appendix - Financial Beta

This appendix is an optional, non-graded finance application. It is not part of the core exercise set.

A1. From Regression Slope to Financial Beta

Financial beta is a regression slope used in finance.

The variables are returns. Let $R_{i,t}$ be the return on asset i in period t, $R_{m,t}$ the market return, and $R_{f,t}$ the risk-free return over the same period.

Financial beta is defined using excess returns:

$$R^e_{i,t}=R_{i,t}-R_{f,t},$$

and

$$R^e_{m,t}=R_{m,t}-R_{f,t}.$$

All returns must be measured over the same horizon. Monthly asset returns require monthly market returns and a monthly risk-free rate. Daily returns require daily quantities.

The beta regression is

$$R^e_{i,t}=\alpha_i+\beta_i R^e_{m,t}+\epsilon_{i,t}.$$

The dependent variable is asset excess return. The explanatory variable is market excess return. The slope β_i measures average sensitivity to market excess returns.

If beta is 1, a one percentage point increase in market excess return is associated, on average, with a one percentage point increase in the asset excess return. If beta is between 0 and 1, the asset moves with the market but less strongly. If beta is greater than 1, the asset is more sensitive than the market. A negative beta means the asset tends to move in the opposite direction.

The covariance formula gives the same slope:

$$\beta_i=\frac{Cov(R^e_i,R^e_m)}{Var(R^e_m)}.$$

The numerator measures co-movement between the asset and market excess returns. The denominator rescales by the variability of the market excess return.

A2. What Beta Measures

Beta measures systematic risk: exposure to broad market movements in excess returns.

It does not measure total risk. An asset can have low beta and still be risky. Company news, sector shocks, liquidity problems, credit events, and other asset-specific forces can all move the asset. In the beta regression, those movements appear in the residual term.

This distinction is important. Two assets can have similar beta values and very different total volatility. They may have similar market exposure but very different residual risk.

Beta is also historical. It is estimated from a sample period. It can change if the company changes, leverage changes, the sector changes, or the market environment changes.

A3. Alpha and Careful Language

In the beta regression, alpha is the intercept:

$$\alpha_i.$$

It is the average excess return not explained by market excess return in this one-factor model.

In investment language, alpha is often used to describe performance beyond market exposure. That interpretation requires evidence. A positive estimated alpha may reflect skill, but it may also reflect noise, missing risk factors, fees, data mining, or a short sample.

Alpha is therefore a claim to test, not something to assume.

A4. Reporting Beta

A professional beta report should be specific.

It should state the asset, the market benchmark, the sample period, the return frequency, whether returns are excess returns, the beta estimate, a standard error or confidence interval, R^2, and any important residual patterns.

The final interpretation should stay close to the model. A beta estimate says how the asset moved with the market in the sample. It does not guarantee future returns, prove skill, or summarize every source of risk.

This is the larger lesson of the lecture. Regression gives a powerful language for relationships between variables. That language is useful only when the question is clear, the model is checked, and the interpretation stays within what the data and design can support.